The Alex Ren Effect: Full-Name Duplication Across American Demographic Groups

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March 2025

Abstract: This paper analyzes the "Alex Ren Effect", a phenomenon of full-name convergence within demographic groups. Leveraging North Carolina's public voter registration database of over 7 million registered voters, we investigate duplication across demographic groups including race, gender, political party, and nativity. Notable findings include higher duplication rates among male voters than female voters (60.12% vs 47.78%), and among US-born voters compared to foreign-born voters (53.09% vs 36.59%). Full name frequency is also shown to closely follow a Zipf-Mandelbrot Law distribution, a novel finding in the field of Onomastics.

1. Preamble

We the Alex Rens, in order to understand why the three of us have the exact same full name, establish something to do over spring break, ensure all registered North Carolina voters are aware that their names and personal data are publicly available, provide the field of Onomastics with its worst-ever paper, promote the pay-per-view Alex Ren battle royale planned this summer, and secure the blessings of everyone we meet asking "Yoooo I know someone else who's also named Alex Ren" to ourselves and all Alex Rens across the United States, do ordain and establish this paper on the Alex Ren Effect.

2. Literature Review

The importance of names, specifically related to demographic signaling, has been extensively studied in Labor Economics. It's been shown that resumes with "White-sounding" names receive 50% more callbacks for interviews compared to identical resumes with "African American-sounding names" (Bertrand & Mullainathan, 2004). Furthermore, minority applicants who "whitened" their resumes by getting rid of racial cues were over twice as likely to get callbacks, even at companies that explicitly claim to value diversity (Kang et al., 2016). The study particularly highlights strategies taken by Asian American applicants, such as changing their first name from "Lei" to "Luke", in an attempt to make them seem more "American".

Focusing specifically on the Asian American experience, significant research has been done on why Asian Americans adopt "Americanized" names and the consequences of that choice. The practice dates back to the 19th century, when discrimination and xenophobia forced Asian immigrants to assimilate as a survival tactic (Yeung, 2021). And, it's been shown that the process of exchanging a Chinese name for an English one and subsequently reverting back represents a significant transition in one's identity, coming with significant psychological costs (Xu, 2018). With Chinese Americans, first names such as Amy and Andrew have been shown to appear much more frequently in the Chinese American population compared to the U.S. populous as a whole; the name Alex is also over 5 times more common among Chinese Americans than the general populous (Bartz, 2009).

A broader analysis of immigrant name changing describes a multitude of motivations behind name changes, such as reducing discrimination, protection against prejudice, simplification of complex names, assuming a new identity, and more (Wayne, 2024). Specifically among Hispanic populations, greater exposure to U.S. culture has been shown to increase the likelihood of giving children English names; this assimilation is more prevalent in daughters compared to sons (Sue & Telles, 2007).

Furthermore, research on the statistical distribution of names is directly relevant to the Alex Ren Effect's mechanism. A study of U.S. and Canadian name distributions found that surname distributions follow a curve of steep decline in frequency; a tiny amount of surnames constitute a significant share of the populations (Tucker, 2001). Chinese surnames, meanwhile, follow an exponential pattern at the top and a power-law pattern at the end.

The specific mathematical law used in this paper, Zipf's law, has been applied in past analysis of first and last names (Baek et al., 2011). However, to our knowledge, it has not been previously applied to full legal names, a core analysis and finding of this paper.

3. Data and Methods

The data leveraged in this paper’s analyses was the Current Voter Registration Data from the North Carolina State Board of Elections (North Carolina State Board of Elections, 2026). This dataset was selected since it is the largest publicly-available dataset that includes both full names and detailed demographic data (race, gender, political party, and nativity).

Racial information within the dataset was separated between race (White, Black, Asian, Native American, Pacific Islander, Mulitiracial, Other) and ethnicity (Hispanic, Non-Hispanic). For our analysis, we combined ethnicity into race, where Hispanic voters are identified as “Hispanic” and Non-Hispanic voters are separated by individual racial categories.

4. Name Duplication

First, we present our findings on name duplication rates across demographics. We note that Asian Americans exhibit a shockingly low duplication rate of 19.63%, compared to 55.00% of all voters. This is interesting given the namesake of this paper; contradictory to past analyses which suggest high concentration rates of certain first names among Asian Americans (Bartz, 2009), we’ve shown the demographic to have a particularly low rate. The comparison of Female and Male duplication rates are also striking; Females have a 47.78% rate, whereas Males have 60.12%. This may suggest a smaller/more concentrated pool of potential male first names among the population surveyed, a result reflected in the top 10 names per demographic of Table 3.

Demographic	**Total Voters**	**Voters With Duplicate Name**	**Duplication Rate**	**Unique Full Names**
Asian	155660	30555	19.63%	134785
Black	1772589	791500	44.65%	1190940
Hispanic	382458	89228	23.33%	319587
Native American	63623	21775	34.23%	48033
Multiracial	35240	1962	5.57%	34192
Other Race	152455	12915	8.47%	144841
Pacific Islander	1017	4	0.39%	1015
Unknown	780257	197268	25.28%	647130
White	5756957	3188655	55.39%	3244793
Female	4461098	2131680	47.78%	2805831
Male	3866145	2324266	60.12%	1991542

Other Gender	773013	197705	25.58%	639452
US-Born	6596518	3502412	53.09%	3834105
Not US-Born	2503738	916038	36.59%	1841224
Democrat	2857881	1221523	42.74%	1948842
Green	4577	27	0.59%	4563
Libertarian	54066	4159	7.69%	51685
Republican	2719946	1331457	48.95%	1710882
Unaffiliated	3463786	1420502	41.01%	2406857
All Voters	9100256	5004854	55.00%	5125616

Table 1: Duplication Rates by Demographic

Regressional Analysis

To best interpret duplication rates between populations of differing sizes, we use an OLS regression between the logarithmic size of a demographic in our dataset and its duplication rate. With this, we predict the duplication rate of each racial demographic, comparing this prediction to the actual value. Racial demographics are used for this regressional analysis, as other factors do not have a sufficient number of categories for meaningful analysis.

Demographic	**Total Voters**	**Actual Duplication Rate**	**Predicted Duplication Rate**	**Residual**
White	5756957	55.39%	44.95%	10.44%
Black	1772589	44.65%	37.91%	6.75%
Hispanic	382458	23.33%	28.73%	-5.40%
Unknown	780257	25.28%	33.00%	-7.72%
Asian	155660	19.63%	23.36%	-3.73%
Other	152455	8.47%	23.23%	-14.76%
Multiracial	35240	5.57%	14.47%	-8.91%
Native American	63623	34.23%	18.01%	16.22%
Pacific Islander	1017	0.39%	-6.73%	7.12%

Table 2: Duplication Rates Regressional Prediction

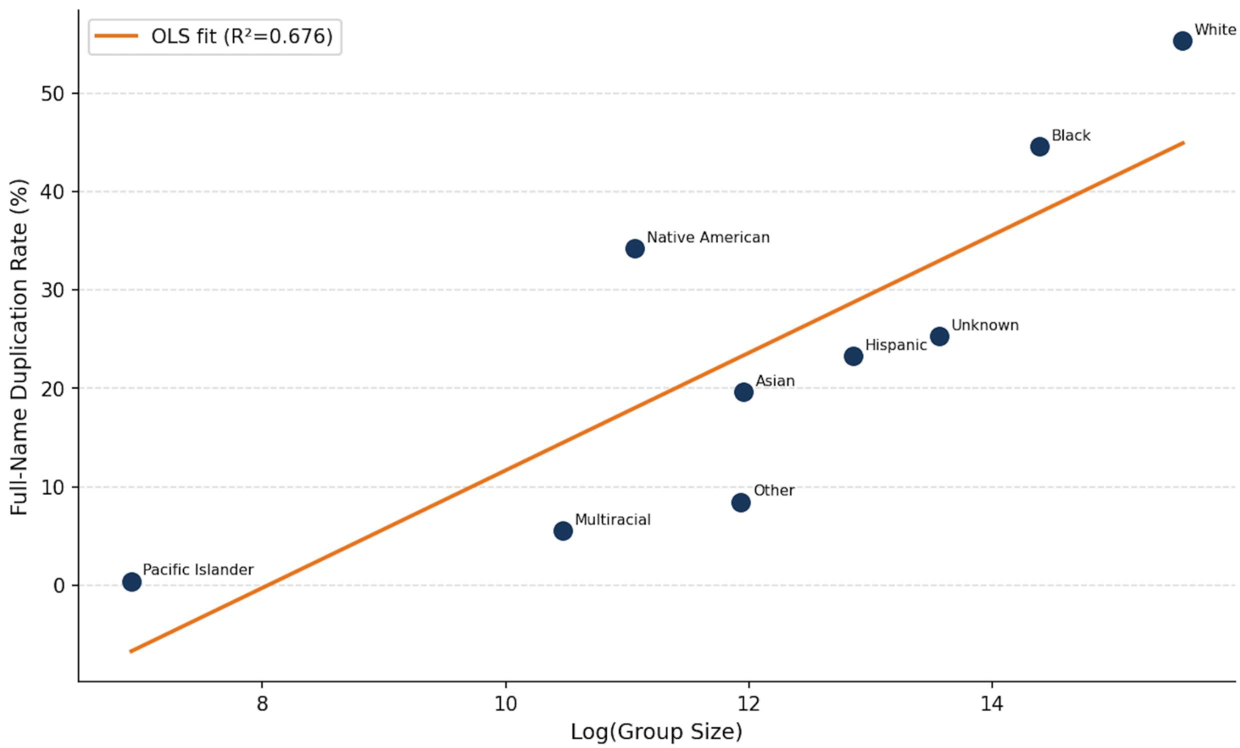

Figure 1: Duplication Rates Regressional Prediction

Comparing the actual and predicted rates, we see that Pacific Islander, Native American, Black, and White demographics observe higher-than-predicted duplication rates, while Multiracial, Other, Asian, Hispanic, and Unknown demographics observe duplication rates below prediction.

5. Zipf's Law

To understand the nature of full-name distributions, we apply Zipf's law and the Zipf-Mandelbrot Law; both are observed in many naturally-occurring frequencies. Formally, Zipf's law is defined as, given a list of ranked items, the frequency f of the element at rank r is given by:

$$f(r) = \frac{C}{r^{\alpha}}$$

Where C is a normalizing constant, and α is the decay component. The Zipf-Mandelbrot Law better fits distributions where higher-ranked elements are more evenly distributed compared to Zipf's law, defined as:

$$f(r) = \frac{C}{(r+b)^{\alpha}}$$

Where b represents a shift parameter.

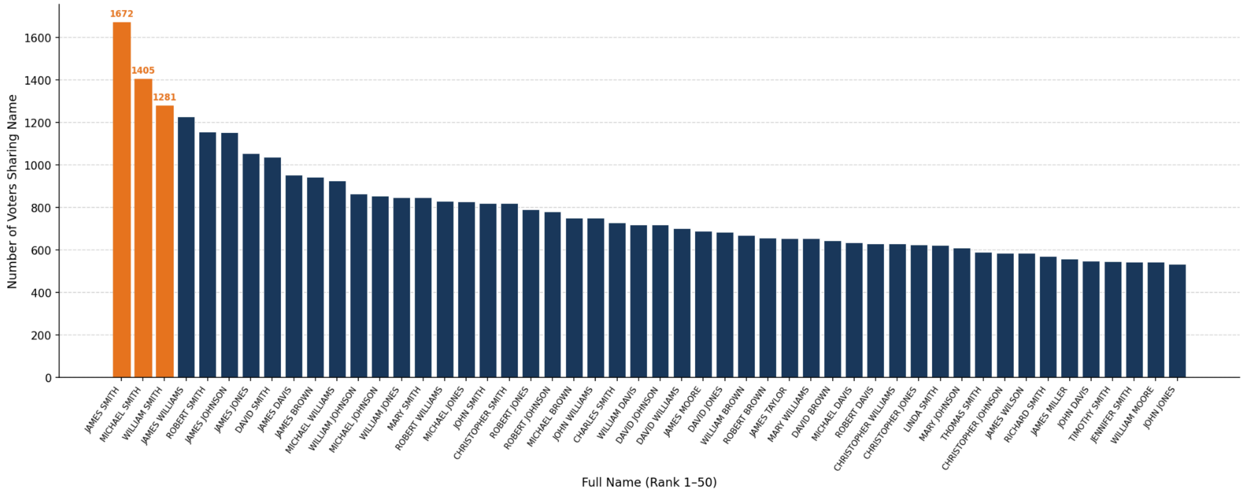

Figure 2: Top 50 Most Common Full Names

Using these laws, we analyze the distribution of full names among the general population. By fitting the α value to our distribution, we closely fit to the Zipf distribution.

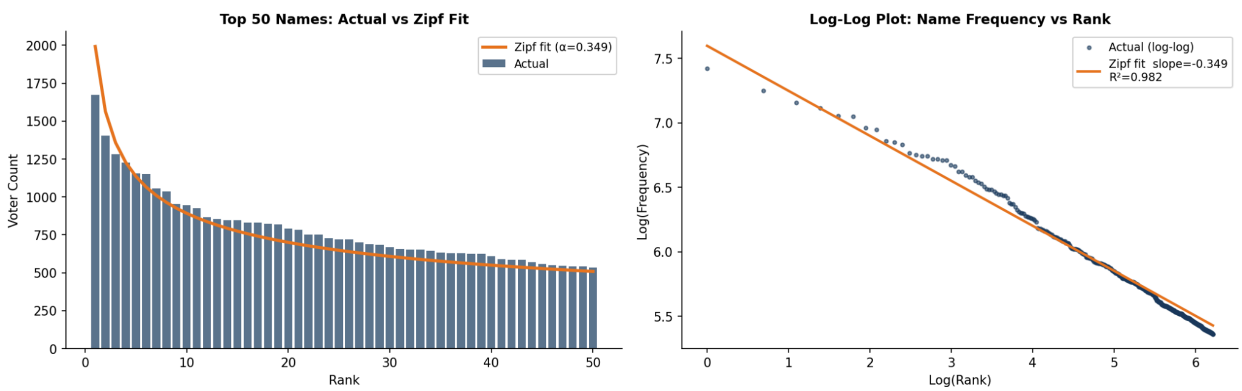

Figure 3: Zipf distribution

However, we notice the leading elements fall behind the predicted graph. As such, we are able to refine our observation by applying the Zipf-Mandelbrot Law:

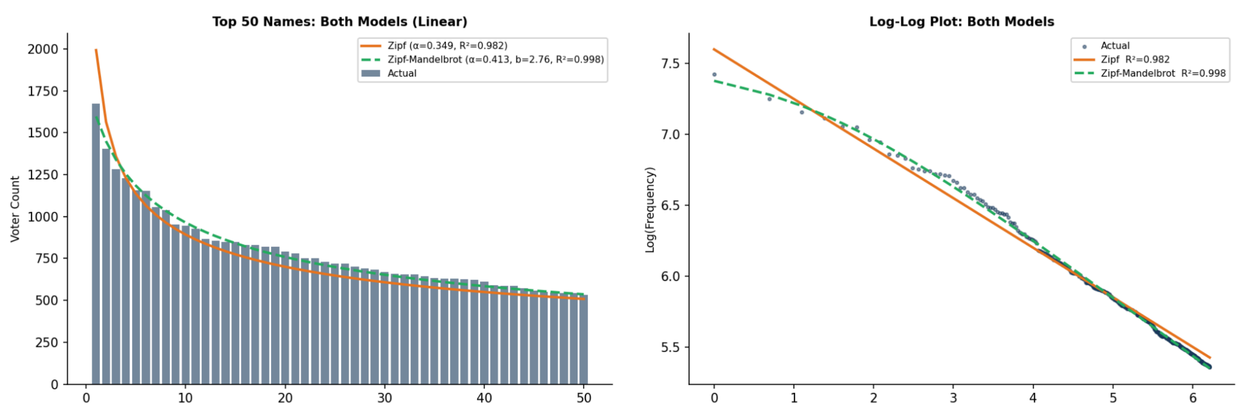

Figure 4: Zipf-Mandelbrot distribution

With this, we observe an astonishing R-squared value of 0.998, showing the full-name distribution of the general population closely matches the Zipf-Mandelbrot distribution.

6. Appendix: Highest-Frequency Names

Below we include the top-ten highest frequency names for each demographic investigated in this paper, along with the frequency of each within our dataset. Highest frequency names overall are presented in Figure 2.

Asian	Black	Hispanic	White	Native American
THANH NGUYEN (67)	JAMES WILLIAMS (470)	JOSE HERNANDEZ (138)	JAMES SMITH (1129)	JAMES LOCKLEAR (130)
PHUONG NGUYEN (57)	JAMES SMITH (362)	JOSE RODRIGUEZ (128)	MICHAEL SMITH (1010)	JAMES HUNT (87)
THUY NGUYEN (51)	JAMES JOHNSON (360)	MARIA RODRIGUEZ (112)	WILLIAM SMITH (992)	MARY LOCKLEAR (67)
MINH NGUYEN (46)	JAMES JONES (330)	MARIA HERNANDEZ (109)	ROBERT SMITH (877)	MICHAEL LOCKLEAR (62)
THAO NGUYEN (43)	MICHAEL WILLIAMS (304)	JOSE MARTINEZ (103)	DAVID SMITH (806)	JOHN LOCKLEAR (58)
TUAN NGUYEN (42)	MARY WILLIAMS (286)	JOSE GARCIA (102)	JAMES JOHNSON (683)	CHRISTOPHER LOCKLEAR (54)
KIM NGUYEN (39)	JAMES BROWN (283)	MARIA GONZALEZ (93)	JAMES WILLIAMS (642)	JAMES OXENDINE (49)
HUNG NGUYEN (37)	MICHAEL SMITH (266)	MARIA MARTINEZ (92)	JAMES DAVIS (611)	WILLIAM LOCKLEAR (49)
TRANG NGUYEN (37)	JAMES DAVIS (265)	JOSE GONZALEZ (90)	MARY SMITH (602)	LINDA LOCKLEAR (47)
YOUNG KIM (36)	ROBERT WILLIAMS (253)	JOSE LOPEZ (86)	WILLIAM JOHNSON (601)	BOBBY LOCKLEAR (44)

Multiracial	Other	Pacific Islander	Unknown
JAMES SMITH (11)	MOHAMED MOHAMED (18)	ANA NICOLE CHARGUALAF (2)	JAMES SMITH (140)
MICHAEL LEE (7)	MOHAMMAD KHAN (16)	THOMAS SMITH (2)	MICHAEL SMITH (111)
JAMES WILLIAMS (6)	JAY PATEL (15)	BRENDA BADUA (1)	MICHAEL WILLIAMS (101)
WILLIAM JOHNSON (6)	AMIT PATEL (14)	JASON COVENEY (1)	JAMES JONES (99)
CHRISTOPHER SMITH (5)	ANH NGUYEN (13)	CERELINA DOLPH (1)	JAMES JOHNSON (97)
CHRISTOPHER WILLIAMS (5)	JAMES SMITH (13)	JAIME GRIFFITH (1)	JAMES WILLIAMS (97)
JAMES JONES (5)	CHIRAG PATEL (13)	MICHAEL HOWERTON (1)	CHRISTOPHER SMITH (82)
ASHLEY JONES (5)	JOSHUA WILLIAMS (13)	ABIGAIL HUNNINGHAKE (1)	JAMES BROWN (80)
WILLIAM SMITH (4)	THUY NGUYEN (13)	EVA JOHNSON (1)	WILLIAM SMITH (80)
TYLER WILLIAMS (4)	PHUONG NGUYEN (13)	WILLIAM KRUEMMEL (1)	MICHAEL JOHNSON (79)

Male	Female	Unknown Gender	US-Born	Not US-Born
JAMES SMITH (1540)	MARY SMITH (812)	JAMES SMITH (129)	JAMES SMITH (1280)	JAMES SMITH (392)
MICHAEL SMITH (1294)	MARY WILLIAMS (619)	MICHAEL SMITH (110)	MICHAEL SMITH (1047)	MICHAEL SMITH (358)
WILLIAM SMITH (1185)	LINDA SMITH (594)	MICHAEL WILLIAMS (97)	WILLIAM SMITH (1013)	JAMES WILLIAMS (293)
JAMES WILLIAMS (1139)	MARY JOHNSON (570)	CHRISTOPHER SMITH (95)	JAMES WILLIAMS (934)	JAMES JOHNSON (283)
ROBERT SMITH (1081)	JENNIFER SMITH (504)	JAMES JOHNSON (94)	ROBERT SMITH (911)	WILLIAM SMITH (268)
JAMES JOHNSON (1058)	MARY JONES (501)	WILLIAM SMITH (92)	JAMES JOHNSON (870)	JAMES JONES (263)
JAMES JONES (965)	PATRICIA SMITH (495)	JAMES JONES (89)	DAVID SMITH (802)	ROBERT SMITH (245)
DAVID SMITH (962)	MARY BROWN (446)	JAMES WILLIAMS (87)	JAMES JONES (793)	DAVID SMITH (236)
JAMES DAVIS (882)	ELIZABETH SMITH (435)	JAMES BROWN (78)	JAMES DAVIS (729)	JAMES BROWN (233)
JAMES BROWN (866)	BARBARA SMITH (422)	MICHAEL JOHNSON (76)	JAMES BROWN (712)	CHRISTOPHER SMITH (226)

Democrat	Republican	Unaffiliated	Libertarian	Green
JAMES SMITH (515)	JAMES SMITH (605)	JAMES SMITH (541)	MICHAEL SMITH (17)	ALYSSA DAVIS (3)
JAMES WILLIAMS (501)	MICHAEL SMITH (543)	MICHAEL SMITH (480)	ROBERT SMITH (11)	LOGAN SMITH (2)
JAMES JOHNSON (414)	WILLIAM SMITH (510)	WILLIAM SMITH (411)	JAMES SMITH (11)	MEGHAN WRIGHT (2)
JAMES JONES (397)	ROBERT SMITH (455)	ROBERT SMITH (404)	CHRISTOPHER SMITH (10)	SHANNON SCOTT (2)
MARY WILLIAMS (372)	DAVID SMITH (434)	JAMES WILLIAMS (376)	MICHAEL JONES (9)	ASHLEY WILLIAMS (2)
MICHAEL SMITH (365)	JAMES JOHNSON (384)	DAVID SMITH (363)	JAMES JOHNSON (9)	TREVOR SHUMATE (2)
WILLIAM SMITH (356)	JAMES DAVIS (346)	JAMES JONES (351)	MATTHEW WILLIAMS (9)	JANIE EKERE (2)
MARY SMITH (337)	JAMES WILLIAMS (344)	MICHAEL WILLIAMS (351)	MICHAEL DAVIS (8)	DANIEL HALL (2)
JAMES BROWN (335)	JAMES BROWN (313)	JAMES JOHNSON (346)	JOSHUA SMITH (8)	MADISON FOSTER (2)
ROBERT WILLIAMS (320)	WILLIAM JONES (312)	MICHAEL JONES (330)	CHRISTOPHER JONES (8)	JESSICA OLSEN (2)

Table 3: Highest Frequency Names by Demographic

7. References

Baek, S. K., Bernhardsson, S., & Minnhagen, P. (2011). Zipf's law unzipped. *New Journal of Physics*, *13*(4), 043004. https://doi.org/10.1088/1367-2630/13/4/043004

Bartz, K. (2009, March 13). *English First Names for Chinese Americans*. Blogs.iq.harvard.edu. https://blogs.iq.harvard.edu/english_first_n

Bertrand, M., & Mullainathan, S. (2004). Are Emily and Greg More Employable than Lakisha and Jamal? a Field Experiment on Labor Market Discrimination. *American Economic Review*, *94*(4), 991–1013. https://doi.org/10.1257/0002828042002561

Kang, S. K., Decelles, K., Tilcsik, A., & Jun, S. (2016, March). *(PDF) Whitened Resumes: Race and Self-Presentation in the Labor Market*. ResearchGate. https://www.researchgate.net/publication/298795100_Whitened_Resumes_Race_and_Self-Presentation_in_the_Labor_Market

North Carolina State Board of Elections. (2026, March 18). *Voter Registration Data | NCSBE*. Www.ncsbe.gov. https://www.ncsbe.gov/results-data/voter-registration-data

Sue, Christina A., & Telles, Edward E. (2007). Assimilation and Gender in Naming. *American Journal of Sociology*, *112*(5), 1383–1415. https://doi.org/10.1086/511801

Tucker, D. K. (2001). Distribution of Forenames, Surnames, and Forename–Surname Pairs in the United States. *Names*, *49*(2), 69–96. https://doi.org/10.1179/nam.2001.49.2.69

Wayne, S. (2024). *Name changing among immigrants | EBSCO*. EBSCO Information Services, Inc. | Www.ebsco.com. https://www.ebsco.com/research-starters/law/name-changing-among-immigrants

Xu, X. (2018). Exploring the Logic of Name Changes and Identity Construction: A Reflective Self-Narration of Assimilation Expectations. *Names*, *68*(1), 32–41. https://doi.org/10.1080/00277738.2018.1452937

Yeung, J. (2021, April 7). *Why some Asian Americans are embracing their heritage by dropping their anglicized names*. CNN. https://www.cnn.com/style/article/asian-american-name-change-hyphenated-intl-hnk-dst

Note: Alex Ren kinda wrote the whole paper, so don't blame Alex Ren or Alex Ren for it.